

Optimal Buy-and-Hold Asset Allocation: A Multi-Horizon Drawdown-Constrained Approach

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Abstract

We investigate optimal buy-and-hold portfolio allocation using drawdown-constrained optimization across multiple investment horizons (5, 10, 20, and 30 years). Using historical ETF data from 1996-2026, we discover a remarkably consistent "40/40/20" framework — 40% growth assets, 40% gold, and 20% Bitcoin — that maximizes risk-adjusted returns across all time horizons. We find that the optimal maximum drawdown constraint converges to approximately -36% to -38% regardless of horizon length, suggesting a fundamental structural property of asset allocation. Our results challenge conventional portfolio wisdom by demonstrating that gold and Bitcoin, traditionally viewed as peripheral allocations, constitute the majority of optimal portfolios when drawdown risk is explicitly constrained. Across all tested horizons, the optimized portfolios deliver Sharpe ratios 2-6× higher than conventional benchmarks including the 60/40 portfolio, the All Weather portfolio, and pure equity strategies.

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